Selected Tree Queries and LDBC SNB SF1

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	D	P	S_D	S_P	B	D	P	S_D	S_P	B	D	P	S_D	S_P
F1K	Q_{desc}	Root	440	2.3	2.3	190x	190x	6.3	3.1	2.5	2.1x	2.5x	30	28	41	1.0x	0.71x
F1K	Q_{desc}	Leaf parent	13	1.9	1.8	6.7x	7.1x	3.6	2.0	2.0	1.7x	1.8x	1.8	1.7	1.9	1.0x	0.95x
WT100	Q_{desc}	Root	26	2.2	1.8	12x	14x	3.5	1.5	1.5	2.3x	2.3x	12	13	14	0.88x	0.86x
WT100	Q_{desc}	Leaf parent	12	1.7	1.7	6.7x	6.8x	3.0	1.5	1.5	2.0x	2.0x	1.7	2.1	1.8	0.82x	0.95x
WT1K	Q_{desc}	Root	1.2K	2.6	2.7	440x	420x	8.7	1.6	2.2	5.3x	3.9x	59	45	47	1.3x	1.3x
WT1K	Q_{desc}	Leaf parent	22	2.3	1.7	9.7x	13x	3.5	1.6	1.7	2.2x	2.0x	2.2	2.3	2.5	0.93x	0.88x
WT10K	Q_{desc}	Root	110K	8.6	11	13Kx	10Kx	25	10	8.6	2.4x	2.9x	410	490	500	0.84x	0.81x
WT10K	Q_{desc}	Leaf parent	110	5.7	1.9	20x	60x	3.0	4.6	2.9	0.65x	1.0x	2.4	2.6	2.6	0.92x	0.93x
WT100K	Q_{desc}	Root	24M	76	74	310Kx	320Kx	120	92	76	1.3x	1.6x	4.3K	5.1K	5.3K	0.84x	0.82x
WT100K	Q_{desc}	Leaf parent	580	37	1.9	15x	300x	2.2	51	27	0.04x	0.08x	1.2	1.3	1.3	0.93x	0.92x
SNB/C	Q_{desc}	Root	92K	680	2.0	140x	47Kx	3.6	420	390	0.01x	0.01x	4.7	4.5	4.7	1.0x	1.0x
SNB/C	Q_{desc}	Leaf parent	22K	720	2.0	31x	11Kx	3.4	440	390	0.01x	0.01x	1.6	1.6	1.6	1.0x	0.99x
SNB/P	Q_{desc}	Root	17K	2.7	2.6	6.2Kx	6.6Kx	6.9	2.5	2.3	2.8x	3.1x	46	41	43	1.1x	1.1x
SNB/P	Q_{desc}	Leaf parent	15K	1.6	1.9	9.0Kx	7.8Kx	3.8	2.1	1.9	1.8x	2.0x	1.1	1.3	1.5	0.87x	0.74x
SNB/T	Q_{desc}	Root	14K	1.9	1.9	7.7Kx	7.6Kx	3.3	1.7	1.6	2.0x	2.1x	9.7	11	11	0.89x	0.85x
SNB/T	Q_{desc}	Leaf parent	14K	1.6	1.7	9.0Kx	8.6Kx	2.9	1.6	1.5	1.9x	2.0x	1.5	1.5	1.5	1.0x	1.0x
F1K	$Q_{a\&d}$	Root–leaf	12	1.4	1.5	8.8x	8.4x	19	1.0	0.91	19x	21x	1.7	1.1	1.6	1.5x	1.1x
F1K	$Q_{a\&d}$	Deep parent–leaf	12	1.6	1.5	7.5x	8.4x	22	1.0	0.92	22x	24x	1.8	1.2	1.6	1.6x	1.2x
WT100	$Q_{a\&d}$	Root–leaf	11	1.7	1.6	6.7x	7.1x	19	0.92	0.94	21x	20x	1.4	0.99	1.1	1.4x	1.3x
WT100	$Q_{a\&d}$	Deep parent–leaf	11	1.5	1.5	7.5x	7.6x	18	0.90	0.94	20x	19x	1.2	1.1	1.3	1.1x	0.98x
WT1K	$Q_{a\&d}$	Root–leaf	13	1.6	1.5	8.1x	8.3x	21	0.84	0.89	25x	23x	1.1	0.81	1.1	1.3x	0.94x
WT1K	$Q_{a\&d}$	Deep parent–leaf	12	1.5	1.5	7.8x	8.0x	19	0.85	0.89	22x	21x	1.3	1.3	1.3	1.0x	0.99x
WT10K	$Q_{a\&d}$	Root–leaf	19	1.7	1.8	11x	11x	21	0.86	0.86	25x	25x	1.4	1.2	0.91	1.2x	1.5x
WT10K	$Q_{a\&d}$	Deep parent–leaf	18	1.8	1.7	11x	11x	21	0.96	0.93	22x	23x	1.2	1.1	1.1	1.1x	1.1x
WT100K	$Q_{a\&d}$	Root–leaf	80	1.9	1.8	42x	44x	20	0.94	0.88	22x	23x	1.6	1.3	1.3	1.2x	1.2x
WT100K	$Q_{a\&d}$	Deep parent–leaf	81	1.8	1.8	44x	45x	23	0.84	0.89	27x	25x	1.3	1.3	1.3	1.0x	0.96x
SNB/C	$Q_{a\&d}$	Root–leaf	15K	2.0	1.9	7.3Kx	7.9Kx	22	1.0	0.93	22x	24x	1.6	1.2	1.1	1.3x	1.4x
SNB/C	$Q_{a\&d}$	Deep parent–leaf	15K	2.1	1.8	6.9Kx	8.0Kx	19	1.3	1.0	15x	19x	1.2	1.1	1.1	1.1x	1.0x
SNB/P	$Q_{a\&d}$	Root–leaf	15K	1.1	1.5	13Kx	9.7Kx	20	0.80	0.89	24x	22x	1.4	1.1	1.1	1.3x	1.3x
SNB/P	$Q_{a\&d}$	Deep parent–leaf	14K	1.6	1.5	9.2Kx	9.7Kx	17	0.88	0.91	20x	19x	1.3	1.1	1.1	1.2x	1.1x
SNB/T	$Q_{a\&d}$	Root–leaf	15K	1.6	1.7	9.0Kx	8.6Kx	19	0.82	0.95	23x	20x	1.2	1.1	1.2	1.1x	1.0x
SNB/T	$Q_{a\&d}$	Deep parent–leaf	14K	1.4	1.7	10Kx	8.6Kx	21	0.83	0.88	25x	23x	1.4	1.1	1.1	1.3x	1.3x
SNB	IC12	–	190K	170K	170K	1.1x	1.1x	400	300	400	1.3x	0.98x	160	120	110	1.3x	1.5x
SNB	IS2	–	61K	3.8K	13K	16x	4.7x	44	240	280	0.18x	0.16x	4.5	4.5	11	0.99x	0.42x
SNB	IS6	–	17K	2.2K	3.3K	8.0x	5.3x	29	23	120	1.3x	0.24x	1.9	2.0	1.8	0.95x	1.0x

Queries: Q_{desc} = All Descendants; $Q_{a\&d}$ = Check if Ancestor; IC12 = Interactive Complex 12; IS2 = Interactive Short 2; IS6 = Interactive Short 6.

Parameters: Root = root of largest tree; Leaf parent = parent of a leaf; Root–leaf = root-farthest-leaf pair; Deep parent–leaf = deepest parent-leaf pair.

Methods: B = Baseline; D = Dewey; P = PrePost; $S_D = B / D$; $S_P = B / P$. All times are in ms.

Graphs: F = forest; WT = ultrawide tree; K = 1,000 nodes; SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes and speedups use about two significant digits; K denotes 1,000 and M denotes 1,000,000.

Speedup color:

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup